www.Padasalai.Net www.TrbTnpsc.com

Tsi12M

Tenkasi District

Common Quarterly Examination - September 2026

21-9-2026.

Standard 12

MATHEMATICS

PART-I

Time Allowed: 3.00 Hours Maximum Marks: 90

Choose the correct answer: 20×1=20

1) If $A\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix} = \begin{bmatrix}6 & 0\\ 0 & 6\end{bmatrix}$, then A =

a) $\begin{bmatrix}1 & -2\\ 1 & 4\end{bmatrix}$ b) $\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$ c) $\begin{bmatrix}4 & 2\\ -1 & 1\end{bmatrix}$ d) $\begin{bmatrix}4 & -1\\ 2 & 1\end{bmatrix}$

2) If $A = \begin{bmatrix}7 & 3\\ 4 & 2\end{bmatrix}$, then $9I_2 - A =$

a) A^{-1} b) $\frac{A^{-1}}{2}$ c) $3A^{-1}$ d) $2A^{-1}$

3) If A^TA^{-1} is symmetric, then $A^2 =$

a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

4) The value of $\sum_{n=1}^{13}\left(i^n + i^{n-1}\right)$ is

a) 1 + i b) i c) 1 d) 0

5) If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is

a) z b) $\bar{z}$ c) $\frac{1}{z}$ d) 1

6) The value of $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$ is

a) $\text{cis}\frac{2\pi}{3}$ b) $\text{cis}\frac{4\pi}{3}$ c) $-\text{cis}\frac{2\pi}{3}$ d) $-\text{cis}\frac{4\pi}{3}$

7) A zero of $x^3 + 64$ is

a) 0 b) 4 c) 4i d) −4

8) If α, β and γ are the zeros of $x^3 + px^2 + qx + r$ then $\sum\frac{1}{\alpha}$ is

a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

9) The number of positive zeros of the polynomial $\sum_{j=0}^{n} nC_r\,(-1)^r x^r$ is

a) 0 b) n c) <n d) r

10) If $\sin^{-1} x = 2\sin^{-1}\alpha$ has a solution, then

a) $|\alpha| \le \frac{1}{\sqrt{2}}$ b) $|\alpha| \ge \frac{1}{\sqrt{2}}$

c) $|\alpha| < \frac{1}{\sqrt{2}}$ d) $|\alpha| > \frac{1}{\sqrt{2}}$

Padasalai.Net

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

11) If $\cot^{-1} x = \frac{2\pi}{5}$ for some $x \in R$, the value of $\tan^{-1}x$ is

a) $-\frac{\pi}{10}$ b) $\frac{\pi}{5}$ c) $\frac{\pi}{10}$ d) $-\frac{\pi}{5}$

12) If $\sin^{-1} x + \cot^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{2}$, then x is equal to

a) $\frac{1}{2}$ b) $\frac{1}{\sqrt{5}}$ c) $\frac{2}{\sqrt{5}}$ d) $\frac{\sqrt{3}}{2}$

13) The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

14) Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

a) 2ab b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

15) The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is

a) $\frac{\sqrt{3}}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{3\sqrt{2}}$ d) $\frac{1}{\sqrt{3}}$

16) If $x + y = k$ is a normal to the parabola $y^2 = 12x$ then the value of k is

a) 3 b) −1 c) 1 d) 9

17) If $\bar{a}$ and $\bar{b}$ are parallel vectors, then $[\bar{a}, \bar{c}, \bar{b}]$ is equal to

a) 2 b) −1 c) 1 d) 0

$c = \pm\sqrt{3}$

18) Distance from the origin to the plane $3x - 6y + 2z + 7 = 0$ is

a) 0 b) 1 c) 2 d) 3

19) If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then

a) $c \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$

20) Find the distance of a point (2,5,−3) from the plane $\vec{r}.\left(6\hat{i} - 3\hat{j} + 2\hat{k}\right) = 5$

a) 1 unit b) 2 units c) 3 units d) 5 units

PART-II

Answer any 7 questions: (Q.no.30 is compulsory) 7×2=14

21) If A is symmetric, Prove that adj A is also symmetric.

22) Prove that $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is orthogonal.

23) Simplify : $i^{1947} + i^{1950}$

24) Solve the equation $x^4 - 9x^2 + 20 = 0$.

25) Find the general equation of the Circle whose diameter is the line segment joining the points (−4,−2) and (1,1).

26) Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$.

27) If $2\hat{i} - \hat{j} + 3\hat{k}$, $3\hat{i} + 2\hat{j} + \hat{k}$, $\hat{i} + m\hat{j} + 4\hat{k}$ are coplanar, find the value of m.

28) Show that the lines $\frac{x-1}{4} = \frac{2-y}{6} = \frac{z-4}{12}$ and $\frac{x-3}{-2} = \frac{y-3}{3} = \frac{5-z}{6}$ are parallel.

Padasalai.Net

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

29) Find the principal value of $\tan^{-1}\left(\sqrt{3}\right)$.

30) The sum of the focal distance of any point on the ellipse is equal to length of the major axis.

PART-III

Answer any 7 of the following questions: (Q.No.40 is Compulsory) 7×3=21

31) Find the adjoint of $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$.

32) Find the rank of $\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$.

33) Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$ into rectangular form.

34) If $|z| = 2$ Show that $3 \le |z + 3 + 4i| \le 7$.

35) Form a polynomial equation with integer coefficients with $\sqrt{\frac{\sqrt{2}}{\sqrt{3}}}$ as a root.

36) Find the value of $\cos^{-1}\left(\cos\frac{\pi}{7}\cos\frac{\pi}{17} - \sin\frac{\pi}{7}\sin\frac{\pi}{17}\right)$.

37) Find the equation of the parabola whose vertex is (5,–2) and focus (2,–2).

38) A concrete bridge is designed as a parabolic arch. The road over bridge is 40m long and the maximum height of the arch is 15m. Write the equation of the Parabolic arch.

39) Find the point of intersection of the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$.

40) State and Prove triangle inequality.

PART-IV

Answer all the questions: 7×5=35

41) a) If $F(\alpha) = \begin{bmatrix} \cos\alpha & 0 & \sin\alpha \\ 0 & 1 & 0 \\ -\sin\alpha & 0 & \cos\alpha \end{bmatrix}$ Show that $[F(\alpha)]^{-1} = F(-\alpha)$.

(OR)

b) Find the foci, vertices and length of major and minor axis of the conic $4x^2 + 36y^2 + 40x - 288y + 532 = 0$.

42) a) If $z = x + iy$ is a complex number such that $\text{Im}\left(\frac{2z+1}{iz+1}\right) = 0$. Show that the locus of z is $2x^2 + 2y^2 + x - 2y = 0$.

(OR)

Padasalai.Net

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

b) If $\vec{a} = 2\vec{i} + 3\vec{j} - \vec{k}$, $\vec{b} = 3\vec{i} + 5\vec{j} + 2\vec{k}$, $\vec{c} = -\vec{i} + 2\vec{j} + 3\vec{k}$, verify that $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

43) a) Solve: $(x - 5)(x - 7)(x + 6)(x + 4) = 504$.

(OR)

b) Solve: $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.

44) a) Draw $y = \sin x$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and $y = \sin^{-1}x$ in $[-1,1]$.

(OR)

b) At a water fountain, water attains a maximum height of 4m at horizontal distance of 0.5m from tis origin. If the Path of water is a parabola, find the height of water at a horizontal distance of 0.75m from the point of origin.

45) a) Solve the following system of linear equation by Gaussian Elimination Method: $2x - 2y + 3z = 2$, $x + 2y - z = 3$, $3x - y + 2z = 1$

(OR)

b) Find the equation of the circle passing through the points $(1,0)$, $(-1,0)$ and $(0,1)$.

46) a) Find the fourth roots of units.

(OR)

b) Prove by vector method that $\sin(\alpha - \beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

47) a) Find the vector (parametric and non-parametric) and Cartesian form of the equations of the plane passing through the three non-collinear points $(3,6,-2)$ $(-1,-2,6)$ and $(6,4,-2)$.

(OR)

b) From the equation whose roots are the squares of the roots of the Cubic equation $x^3 + ax^2 + bx + c = 0$.

Padasalai.Net

Kindly Send Me Your Study Materials To Us Email ID: padasalai.net@gmail.com